Module 15: Delta Boost Audit Certificate

Paper Section: 'The Exceptional Prime Set for pi/10' | Audit Date: May 2026

Battle Plan v1.6 -- David Fox

This certificate audits all numerical claims in the attached LaTeX paper section. Four independent errors were found. One claim (S_4 membership) is correct. All correct values are computed here at 100 dps and SHA-bound.

Audit Summary

<table>
<tr><th>Claim</th><th>Paper States</th><th>TRUE Value</th><th>Verdict</th></tr>
<tr><td>C1: Table ||p*pi/10||</td><td>16 entries (all wrong)</td><td>See Section 2</td><td>4 ERRORS*</td></tr>
<tr><td>C2: delta_p > 0 (Lemma)</td><td>Holds for p in S_4</td><td>CONFIRMED</td><td>PASS</td></tr>
<tr><td>C3: Individual delta_p</td><td>E.g. delta_191=11.719</td><td>delta_191=0.169</td><td>WRONG (E1+E2)</td></tr>
<tr><td>C4: Delta_DS^(4)=23.797</td><td>23.796910 +- 1e-6</td><td>2.753126...</td><td>WRONG (E1+E2)</td></tr>
<tr><td>C5: p_5 > 6*10^12</td><td>min(S\S_4) > 6e12</td><td>p_5=3.994e12</td><td>WRONG (E4)</td></tr>
<tr><td>C6: S_4={2,3,19,191}</td><td>{2,3,19,191}</td><td>{2,3,19,191}</td><td>CORRECT</td></tr>
</table>

* C1 has 16 wrong values; C3 and C4 each carry two compounding errors (E1+E2).

Error Descriptions

<table>
<tr><th>ID</th><th>Location</th><th>Description</th></tr>
<tr><td>E1</td><td>Table (Thm 2)</td><td>All 16 ||p*pi/10|| values are wrong. Errors from 1.79e-5 (p=2) to 3.76e-1 (p=43). The SHA 'a7f3d1b9e5c2...' is truncated and unverifiable. AR</td></tr>
<tr><td>E2</td><td>Thm 4 proof</td><td>Sign error in delta_p computation. DEFINITION: delta_p = -log(||.||) - log(p). PAPER COMPUTES: delta_p = -log(||.||) + log(p) [adds log(p) ir</td></tr>
<tr><td>E3</td><td>Thm 4</td><td>Delta_DS^(4) = 23.796910 follows from E1 + E2 compounding. Correct value: 2.753126... (off by factor ~8.6).</td></tr>
<tr><td>E4</td><td>Rem 2 (S14)</td><td>p_5 > 6*10^12 is wrong. Certified p_5 = 3,993,746,143,633 = 3.994*10^12 < 6*10^12. M4 proves p_5 > 10^10 (completeness bound); the ex</td></tr>
</table>

Section 2: Corrected Table of ||p*pi/10|| Values

Computed at 100 dps via mpmath 1.3.0. Natural log throughout. Formula: ||p*pi/10|| = min(frac(p*pi/10), 1 - frac(p*pi/10)).

<table>
<tr><th>p</th><th>Paper (wrong)</th><th>TRUE ||p*pi/10||</th><th>Err</th><th>1/p</th><th>Hit?</th><th>Note</th></tr>
<tr><td>2</td><td>0.37166359</td><td>0.3716814693</td><td>1.8e-05</td><td>0.50000000</td><td>YES</td><td>S_4</td></tr>
<tr><td>3</td><td>0.05759539</td><td>0.0575222039</td><td>7.3e-05</td><td>0.33333333</td><td>YES</td><td>S_4</td></tr>
<tr><td>5</td><td>0.42925878</td><td>0.4292036732</td><td>5.5e-05</td><td>0.20000000</td><td>no</td><td></td></tr>
<tr><td>7</td><td>0.20092865</td><td>0.1991148575</td><td>1.8e-03</td><td>0.14285714</td><td>no</td><td></td></tr>
<tr><td>11</td><td>0.45610154</td><td>0.4557519189</td><td>3.5e-04</td><td>0.09090909</td><td>no</td><td></td></tr>
<tr><td>13</td><td>0.08443515</td><td>0.0840704497</td><td>3.6e-04</td><td>0.07692308</td><td>no</td><td></td></tr>
<tr><td>17</td><td>0.34110602</td><td>0.3407075111</td><td>4.0e-04</td><td>0.05882353</td><td>no</td><td></td></tr>
<tr><td>19</td><td>0.03027262</td><td>0.0309739582</td><td>7.0e-04</td><td>0.05263158</td><td>YES</td><td>S_4</td></tr>
<tr><td>23</td><td>0.28694349</td><td>0.2256631033</td><td>6.1e-02</td><td>0.04347826</td><td>no</td><td></td></tr>
<tr><td>29</td><td>0.45839086</td><td>0.1106186954</td><td>3.5e-01</td><td>0.03448276</td><td>no</td><td></td></tr>
<tr><td>31</td><td>0.08672446</td><td>0.2610627739</td><td>1.7e-01</td><td>0.03225806</td><td>no</td><td></td></tr>
<tr><td>37</td><td>0.25817184</td><td>0.3761071817</td><td>1.2e-01</td><td>0.02702703</td><td>no</td><td></td></tr>
<tr><td>41</td><td>0.48650271</td><td>0.1194701203</td><td>3.7e-01</td><td>0.02439024</td><td>no</td><td></td></tr>
<tr><td>43</td><td>0.11483631</td><td>0.4911515896</td><td>3.8e-01</td><td>0.02325581</td><td>no</td><td></td></tr>
<tr><td>191</td><td>0.00155435</td><td>0.0044196836</td><td>2.9e-03</td><td>0.00523560</td><td>YES</td><td>S_4</td></tr>
<tr><td>197</td><td>0.11343472</td><td>0.1106247243</td><td>2.8e-03</td><td>0.00507614</td><td>no</td><td></td></tr>
</table>

Qualitative conclusion (which primes are in S_4) is CORRECT in the paper despite all 16 ||.|| values being wrong.

Section 3: Corrected delta_p Values and Sum

Definition (from paper): delta_p = -log(||p*pi/10||) - log(p) [natural log]. Correct computation uses TRUE ||p*pi/10|| values and the correct minus sign.

p	Paper delta_p	TRUE ‖p*pi/10‖	TRUE -log(‖.‖)	TRUE log(p)	TRUE delta_p	Status
2	1.682791	0.3716814693	0.989718	0.693147	0.29657088	CORRECT
3	3.953040	0.0575222039	2.855584	1.098612	1.75697196	CORRECT
19	6.442201	0.0309739582	3.474608	2.944439	0.53016951	CORRECT
191	11.718878	0.0044196836	5.421687	5.252273	0.16941375	CORRECT
SUM	**23.796910**	**--**	**--**	**--**	**2.75312609**	**CORRECT**

TRUE Delta_DS^(4) = 2.7531260943232951007 vs paper claim 23.796910 (off by 21.0438, factor ~8.6x)

Section 4: p_5 Bound

Item	Paper Claims	Certified Value	Source	Verdict
p_5 lower bound	> 6 * 10^12	p_5 = 3,993,746,143,633 = 3.994*10^12	M4 SHA b810a7a3...	WRONG
S_4 completeness	cap [2, 10^10]	p_4 = 191 < 10^4; p_5 > 10^10	M3 SHA e687bb09...	CORRECT

Chain of Custody

Item	SHA-256
m15_delta_boost.py (script)	055b24376f1af17ccc4ed7370a3b7d74823ef279778f9137...
m15.out (stdout)	cf1620c7b8d8b931fe4ceb051b0db9ab20aaa1e3f439929d...
Parent M3 (CF pi/10)	e687bb0931f2c37c6ae12cfbde7ff9a79b3cccf5b0c88e17...
Parent M4 (S_14, p_5)	b810a7a331e47066e3eb4765a5ffdc17c1a56ddbff855a09...

Conclusion

The paper section contains 4 independent numerical errors: wrong table values (E1), a sign error in delta_p (E2), a wrong Delta_DS^(4) sum (E3), and a wrong p_5 bound (E4). The only fully correct claim is S_4 = {2,3,19,191}. Corrected certified values: delta_2=0.29657, delta_3=1.75697, delta_19=0.53017, delta_191=0.16941; Delta_DS^(4)=2.75313; p_5=3993746143633. All values computed at 100 dps and SHA-bound in m15.out.

CERTIFIED -- Battle Plan v1.6 -- David Fox -- May 2026